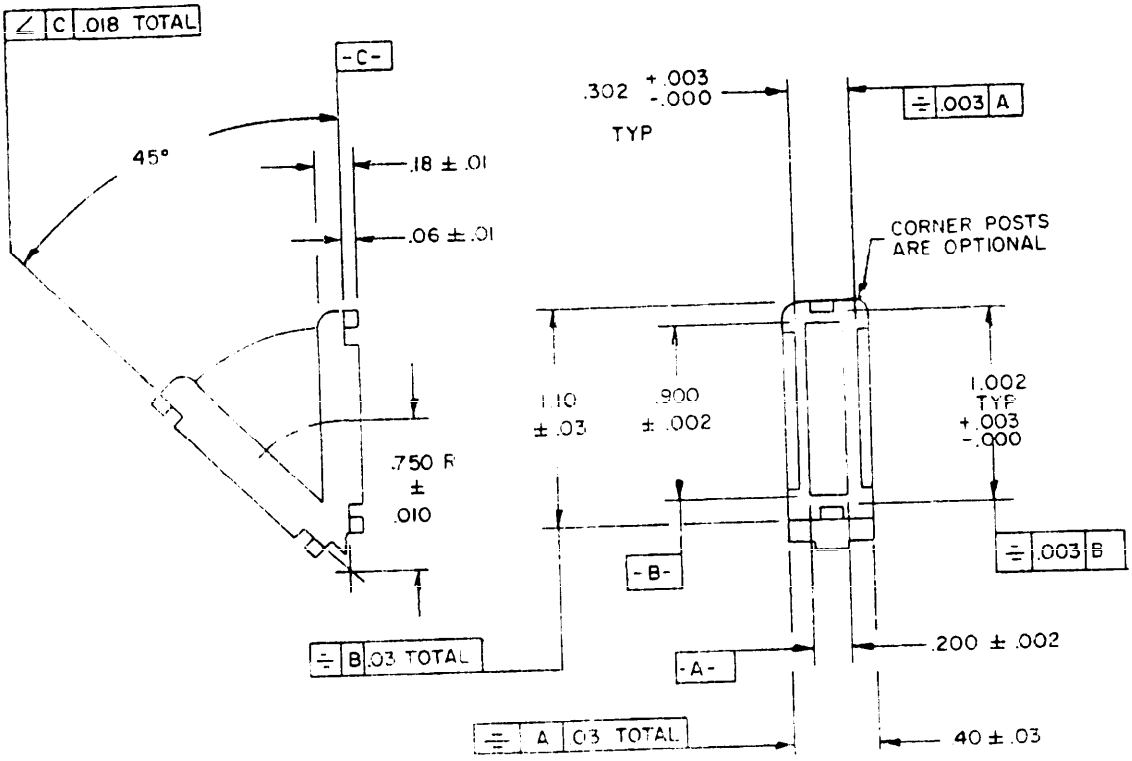
MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLY, RIGID, CAST, 45-DEGREE H-PLANE BEND, REDUCED HEIGHT, FULL CORRAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assembly described herein shall consist of this document and the latest issue of Specification MIL-W-3970.



INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.03	.8	.40	10.2
.003	.08	.06	1.5	.750	19.05
.010	.25	.18	4.6	.900	22.86
.01	.3	.200	5.08	1.002	25.45
.018	.46	.302	7.67	1.10	27.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Dimensions for reduced height and full corral configuration.

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REQUIREMENTS:

Physical characteristics:
Dimensions and configuration: See figure 1.
Operating temperature range: -55° to +300°C.
Material: D712.0, temper F, aluminum alloy per QQ-A-601 or C712.1 per QQ-A-371.
Surfaces (sandblasting after machining permissible):
Internal: $63\sqrt{}$ microinches per ANSI B46.1.
External: $125\sqrt{}$ microinches per ANSI B46.1.

Fungus resistance: Materials nutrient to fungus shall not be used.

Pressurization: 45 lb_f/in^2 gage maximum.

Penetrant inspection: No indication of cracks when tested per MIL-I-6866, type I, method B.

Electrical characteristics:
Frequency range: 8.2 to 12.4 GHz.
VSWR: 1.05:1 maximum.
Power:
Peak - 50 kW.
Average - 10 kW.

Part number: M3970/30-001.

NOTE: The mating waveguides are 1/2 height of X-band waveguides, with a nominal wall thickness of .050 inch.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-0881-10)

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